Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/14/2006
PI Name:	Kostelecky, V. Alan Ph.D.		
Project Title:	Theoretical Studies of Lorentz and CPT Symmetry		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47405	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2914		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		
Task Description:	A general extension of the Standard Model of particle physics incorporating gravity, known as the Standard-Model Extension (SME), allows for the possibility of Lorentz and CPT violation while otherwise preserving conventional physics. Sensitive experiments searching for predicted effects include ones that may be advantageously performed in a microgravity environment. To develop and study theoretically possible violations of spacetime symmetry and to identify sensitive experimental tests for the associated effects. High-sensitivity tests of fundamental symmetries can shed light on physics at the Planck scale and could offer experimental support for an underlying theory such as strings.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	Back in 2004, the structure of the gravitational sector of the SME was published. As a further development of this earlier paper, an investigation of the massless Nambu-Goldstone modes arising from spontaneous Lorentz violation was performed this year. The new work has a variety of implications, including the striking one that the photon may be one observable result of Lorentz violation. This research showed that up to 10 Lorentz and diffeomorphism Nambu-Goldstone modes can appear and that they are contained within the 10 modes of the vierbein associated with gauge degrees of freedom in a Lorentz-invariant theory. A general treatment of spontaneous local Lorentz and diffeomorphism violation was presented in the published paper for various spacetimes. The results are illustrated within the general class of bumblebee models involving vacuum values for a vector field. In Minkowski and Riemann spacetimes, the bumblebee model provides a dynamical theory generating a photon as a Nambu-Goldstone boson for spontaneous Lorentz violation. Associated effects of potential experimental relevance include Lorentz-violating couplings in the matter and gravitational sectors of the Standard-Model Extension and unconventional Lorentz-invariant couplings. In Riemann-Cartan spacetime, the possibility also exists of a Higgs mechanism for the spin connection, leading to the absorption of the propagating Nambu-Goldstone modes into the torsion component of the gravitational field. A second work related to oscillations of the Lorentz-violating fields addressed the extent to which Lorentz violation is generic or even ubiquitous in prospective fundamental theories. The study of this issue was initiated in the context of the elegant possibility that Lorentz symmetry is spontaneously broken in the underlying theory. The published paper performs an investigation of gauge-noninvariant vector field theories with superficially nonrenormalizable nonpolynomial interactions. A primary result was that there exists a large class of nontrivial relevant and stable theories that are also asymptotically free and therefore of theoretical interest. Again, in this case, the Nambu-Goldstone modes of these theories can be identified with the photon, with potential experimental implications. Another direction of research this year concerns the issue of the masslessness of gravitons. In general relativity, gravitational waves propagate at the speed of light, and gravitons are massless. The usual explanation for this is associated with the symmetry of gravity under diffeomorphisms. The possibility was investigated that masslessness can instead arise from spontaneous violation of local Lorentz invariance. The corresponding theory of gravity was constructed. It reproduces the Einstein-Hilbert action of general relativity at low energies and temperatures. Detectable signals occur for sensitive experiments, including some that could be sought in space-based experiments. This work won the Gravity Research Foundation Third Award in 2005.

Bibliography Type:	Description: (Last Updated: 03/05/2008)
Articles in Peer-reviewed Journals	Bluhm R, Kostelecky VA. "Spontaneous Lorentz violation, Nambu-Goldstone modes, and gravity." Physical Review D 2005 Mar;71(6):065008. http://dx.doi.org/10.1103/PhysRevD.71.065008 , Mar-2005
Articles in Peer-reviewed Journals	Altschul B, Kostelecky VA. "Spontaneous Lorentz violation and nonpolynomial interactions." Physics Letters B, 2005 Nov;628(1-2, 3): 106-12. http://dx.doi.org/10.1016/j.physletb.2005.09.018 , Nov-2005
Articles in Peer-reviewed Journals	Kostelecky VA, Potting R. "Gravity from local Lorentz violation." General Relativity and Gravitation 2005 Oct;37(10):1675-9. http://dx.doi.org/10.1007/s10714-005-0149-1 , Oct-2005
Books/Book Chapters	Kostelecky VA, editor. "Proceedings of the Third Meeting on CPT and Lorentz Symmetry, Bloomington, USA, 4-7 August 2004." Hackensack, N.J. : World Scientific, c2005., Jan-2005